

Experiment 5: DBMS

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1. Aim of the Session

To demonstrate the creation of a database schema, population of records, and the use of the CASE Expression to perform conditional analysis on numerical data (parity check).

2. Objective of the Session

- To learn how to manage table lifecycles using DROP and CREATE.
- To understand the application of the modulo operator (%) in SQL.
- To gain proficiency in creating virtual columns using CASE-WHEN-THEN-ELSE logic.
- To practice basic Data Manipulation Language (DML) and Data Query Language (DQL) operations.

3. Practical / Experiment Steps

The experiment consists of the following logic:

1. Table Setup: Ensure a clean environment by removing any existing employee table and creating a new one with specific constraints (Primary Key).
2. Data Population: Insert six distinct records representing employee IDs, names, and salaries.
3. Conditional Querying: Execute a SELECT statement that calculates the remainder of the salary divided by 2 (salary%2).
4. Classification: If the remainder is 0, classify as 'Even Salary'; otherwise, classify as 'Odd Salary' using a CASE statement.

4. Procedure of the Practical

(i) Start your SQL environment (e.g., MySQL, PostgreSQL, or SQL Server Management Studio).

- (ii) Open a new Query Editor window.
- (iii) Create or select the required database schema.
- (iv) Type the DROP and CREATE commands to define the table structure.
- (v) Execute the INSERT commands to populate the table with the provided dataset.
- (vi) Write the SELECT query incorporating the CASE expression for salary classification.
- (vii) Execute the script and verify that the salary_type column correctly reflects the parity of the salary.
- (viii) Note down the results and take a screenshot of the output grid.

5. I/O Analysis (Input / Output Analysis)

Input: Employee Table Data

emp_id	emp_name	salary
101	ayush	50024
102	akash	25001
103	bimlesh	45120
104	chandan	48203
105	dinesh	68103
106	kashish	79536

Output: Query Results

emp_id	emp_name	salary	salary_type
101	ayush	50024	Even Salary
102	akash	25001	Odd Salary
103	bimlesh	45120	Even Salary
104	chandan	48203	Odd Salary
105	dinesh	68103	Odd Salary

emp_id	emp_name	salary	salary_type
106	kashish	79536	Even Salary

6. Output Screenshot:

	emp_id [PK] integer	emp_name character varying (50)	salary integer	salary_type text
1	101	ayush	50024	Even Salary
2	102	akash	25001	Odd Salary
3	103	bimlesh	45120	Even Salary
4	104	chandan	48203	Odd Salary
5	105	dinesh	68103	Odd Salary
6	106	kashish	79536	Even Salary

7. Learning Outcome

Through this practical session, I have:

- Concepts Understood: Mastered the CASE statement syntax, which allows for "if-then-else" logic directly within a SQL result set.
- Skills Developed: Gained the ability to categorize data dynamically without altering the physical storage of the table.
- Practical Exposure: Learned how to handle table resets (Drop/Create) and perform mathematical operations like modulo within a query to derive meaningful business insights.